SimpleChef: A Mobile Cooking Assistant
HCI 4063/5063 Project Proposal

SimpleChef Team

I. Introduction

SimpleChef is a new mobile-first cooking assistant application designed to reduce friction across the entire cooking process—from discovering a recipe to meal planning and grocery shopping. The tagline "Plan. Cook. Shop. Simplify." captures the app's goal: to serve as a capable, calm kitchen companion that streamlines daily cooking tasks.

Existing cooking applications often fragment the experience. Users may browse recipes in one app, track calories in another, and maintain grocery lists in notes or paper. Recipe instructions are frequently buried in long blocks of text, making it difficult to follow steps while cooking. Timer management during cooking can be cumbersome when users must switch between apps or rely on separate kitchen timers. SimpleChef addresses these pain points by unifying recipe discovery, step-by-step cooking mode with integrated timers, meal planning with calorie tracking, and an auto-generating grocery list into a single, cohesive interface.

The application targets home cooks who want to cook more often but find the logistics—planning meals, tracking ingredients, following instructions—overwhelming. By applying Human-Computer Interaction (HCI) principles such as progressive disclosure, clarity over clutter, and smart defaults with full user control, this proposal outlines the design and implementation of SimpleChef as an HCI project.

II. Issues with the Existing UI / Problem Statement (For New Designs)

As a new design concept, SimpleChef addresses several recurring usability and workflow issues observed in existing cooking and meal-planning applications:

- **Fragmented User Experience**: Users must often switch between multiple applications for recipes, meal planning, calorie tracking, and grocery lists. Context switching increases cognitive load and reduces the likelihood that users will maintain consistent meal planning or calorie tracking habits.
- **Poor Recipe Instruction Presentation**: Many recipe apps display instructions as long, uninterrupted paragraphs. During active cooking, users need scannable, step-by-step guidance with ingredients clearly associated with each step. Without "mise en place" (ingredients per step) visibility, users may miss ingredients or lose their place.
- **Ineffective Timer Integration**: Cooking often requires multiple timers (e.g., for pasta, sauce, and resting). Existing apps rarely integrate timers directly into recipe steps, forcing users to leave the app, use separate devices, or manually set external timers. This breaks flow and increases the risk of errors.
- **Manual Grocery List Management**: Building a grocery list from planned meals is tedious when done manually. Users must read through multiple recipes, aggregate ingredients, and deduplicate. Errors and omissions are common, especially when scaling recipes or adjusting servings.
- **Recipe Input Friction**: Adding recipes from text, images, or video sources typically requires manual retyping. Users who find recipes online or in cookbooks face high barriers to importing them into a structured format, which discourages adoption of digital recipe management.
- **Inflexible Calorie and Diet Tracking**: Calorie tracking in generic apps does not integrate with recipes or meal plans. Users cannot easily see how planned meals contribute to daily goals or filter recipes by dietary restrictions without manual cross-referencing.

These problems form the problem statement that SimpleChef aims to solve through an integrated, user-centered design.

III. Project Goals

The primary goals of this project are to:

- **Unify the Cooking Workflow**: Integrate recipe browsing, step-by-step cooking mode, meal planning, calorie tracking, and grocery list generation into a single, coherent application with consistent navigation and data flow.
- **Optimize the Step-by-Step Cooking Experience**: Design a cooking mode that displays only the relevant step, its associated ingredients (mise en place), and embedded timer triggers. The interface must remain legible and calm during active cooking, with minimal cognitive load.
- **Implement Intelligent Timer Management**: Provide a timer dock that displays active timers (stacked by least time remaining), expands on tap for full control, and integrates seamlessly into recipe steps so users never leave the cooking context.
- **Automate Grocery List Generation**: Auto-populate grocery lists from planned meals, deduplicate and categorize items, and allow full user control to edit, reorder, and export lists to notes or external apps.
- **Reduce Recipe Input Friction**: Support multiple input methods—paste text/URL, upload image, upload video, or manual entry—with AI-assisted parsing to structure

ingredients and steps. Provide a verification and edit flow so users can correct parsing errors before saving.

- **Support Meal Planning and Calorie Goals**: Enable users to assign recipes to calendar days, view daily calorie totals against profile goals, and filter recipes by dietary restrictions. Provide optional statistics for trends over time.
- **Apply HCI Principles**: Design for clarity over clutter, progressive disclosure, and visual warmth. Validate the design through user testing and iteration to ensure usability, responsiveness, and aesthetic appeal.

IV. Proposed UI

The proposed SimpleChef UI is designed around five main screens accessible via a bottom navigation bar: Home (Recipe Library), Calendar (Meal Planning), Add Recipe, Grocery List, and Profile. A global header with a back button appears on all sub-screens. The design follows a warm, appetizing color palette (primary: Sea Green #2E8B57; accent: Sandy Yellow #F4A261) and uses a legible sans-serif typography for clarity during cooking.

Key Features:

- **Home Screen**: Responsive grid of recipe cards (2–3 columns) with hero images, titles, and badges (e.g., cook time, dietary tags). Search and filter by dietary restrictions, cook time, and difficulty.
- **Recipe Detail**: Scrollable view with hero image, key info strip (time, difficulty, servings, calories), expandable ingredients, and a prominent "Begin Cooking" button.
- **Step-by-Step Cooking Mode**: Fixed top bar with recipe name, step progress, and a timer dock showing stacked active timers. Each step displays a "mise en place" checklist of ingredients for that step, followed by instructions with embedded timer triggers. The bottom nav is replaced with cooking-specific controls (prev/next step, view all ingredients).
- **Calendar & Meal Planning**: Monthly calendar with colored dots for planned days. Tapping a day reveals a bottom panel for adding meals from the library or quick-add foods, with daily calorie totals and optional statistics view.
- **Grocery List**: Auto-generated from meal plans, grouped by user-editable categories. Items are editable, draggable, and exportable via share sheet.
- **Add Recipe**: Input methods include paste text/URL, upload image, upload video, or manual entry. AI parsing populates structured fields; users verify and edit before saving.
- **Profile**: User info, calorie goals, dietary restrictions for filtering, and friends list for sharing recipes.

Tools and Technology:

- **Frontend**: React Native (Expo) for cross-platform mobile development.
- **Backend**: FastAPI (Python) with PostgreSQL database.
- **Design**: Figma for wireframes and high-fidelity prototyping.
- **AI/ML**: For recipe parsing from text, image, or video inputs (to be integrated).

Figure 1 shows a wireframe of the main home screen layout, including the navigation bar and recipe grid.

Fig. 1. Wireframe of the SimpleChef home screen (Recipe Library) with bottom navigation.

V. Project Timeline

The project will follow the course milestones and an internal task breakdown:

- **Project Proposal (Current)**: Define scope, complete proposal document with wireframe, and finalize design specifications.
- **Weeks 1–4 (Through March 26)**: Core prototype development. Implement recipe library, step-by-step cooking mode with timer dock, and meal planner calendar. Conduct initial user feedback. Deliver Progress Report with GANTT chart, prototype demo, and documentation.
- **Weeks 5–7 (March 27–April 28)**: Complete grocery list automation, recipe input flow (manual and AI parsing), profile and settings. User testing and iteration. Finalize functional prototype. Project Delivery due April 28.
- **Final Demo (Starting April 9)**: In-class presentation with live demonstration of the prototype, summary of design decisions, and UX improvements.
- **April 28–May 3**: Final documentation, system block diagram, design rationale, and user analysis results. Final Project Report due May 3.

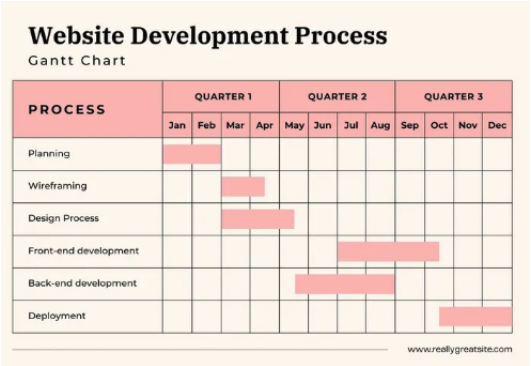

Fig. 2. Project Timeline (GANTT Chart)

VI. Conclusion

In conclusion, SimpleChef proposes a new mobile cooking assistant that addresses fragmentation, poor instruction presentation, and manual grocery management in existing solutions. By unifying recipe discovery, step-by-step cooking mode with integrated timers, meal planning, calorie tracking, and auto-generating grocery lists, the application aims to reduce friction and support users in cooking more consistently. The design prioritizes clarity, progressive disclosure, and user control, with the step-by-step cooking mode as the core experience. Achieving the outlined goals will result in a functional prototype that demonstrates the application of HCI principles to a real-world problem, with opportunities for user testing and iterative refinement throughout the project lifecycle.

References

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